SDN Framework for Naval Applications

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***Abstract*—Abstract– This project implements a Software Defined Networking (SDN) and Software Defined Radio (SDR) framework for improving communication resilience and performance within a simulated U.S. Navy Carrier Strike Group (CSG). Simulated in MATLAB, the framework models dynamically routed traffic between 5G and SATCOM links based on mission, distance, and environmental conditions. Tactical traffic is prioritized for low latency 5G links, while administrative traffic leverages SATCOM for reliability. To evaluate performance under degraded conditions, the simulation introduces periodic link failures of both 5G and SATCOM. Across 100 simulated discrete events, the SDN controller dynamically adapts routing decisions to maintain connectivity and reliability. Results show that 5G achieves lower latency (∼60-85 ms) compared to SATCOM (∼480-620 ms), while SDN redundancy maintains high reliability even under link degradation. These results demonstrate that SDN controlled communication architectures significantly improve adaptability, latency performance, and resilience in maritime environments.**

I. INTRODUCTION

Modern Carrier Strike Groups (CSGs) operate in environments where continuous, reliable, and resilient communications are critical to mission success [1]. A typical CSG includes a nuclear powered aircraft carrier, multiple guided missile destroyers, and a supply or logistics ship, all of which must coordinate across wide geographic areas and dynamic threat environments [2]. We recognize other assets which may be a part of a CSG but for the purposes of our simulation, these will be simplified. These ships rely on a mix of HF, VHF, UHF, 5G, and SATCOM links to exchange sensor data, command and control (C2) messages, and logistics information [3], [4]. However, traditional communication architectures are largely hardware centric and statically configured, making it difficult to adapt to changing conditions such as degraded line of sight, adverse weather, jamming, and fluctuating bandwidth demands [1], [5].

Conventional naval networks also lack a unified mechanism for intelligently prioritizing traffic across the strike group [2]. A destroyer conducting air defense requires higher assurance and lower latency than a supply ship transmitting routine logistics data, yet both often share the same physical channels and routing mechanisms [3]. When weather, distance, or interference erode the performance of a given link, the network may not automatically reroute traffic over alternative paths such as multi hop ship-carrier-ship relays or SATCOM based backhaul. Satellite backhauls serve to transmit data between remote locations and the core network [4]. This rigidity limits the strike group's ability to maintain assured communications in contested or congested electromagnetic environments [5].

Recent developments in Software Defined Networking (SDN) offer a promising path forward [7], [8]. An SDN separates the control and data planes, enabling a logically centralized controller to manage flows, enforce policies, and react to changing network conditions in real time [9]. The network can observe link performance and mission priorities, then dynamically choose which bands and paths to use [10], [11].

This project applies the above concept to a simulated CSG modeled in Matrix Laboratory (MATLAB) [11]. The simulated scenario, shown in Fig. 1, consists of five ships (only two are shown to maintain image clarity), one aircraft carrier, a satellite relay, and a shore node.

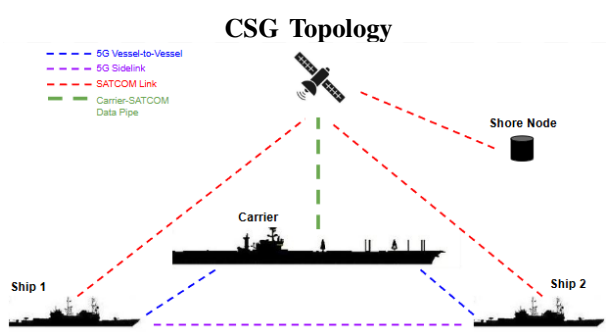

Fig. 1. Topology of simulated CSG

Within this framework, the aircraft carrier functions as an SDN master controller that selects communication paths between 5G and SATCOM links. The decision framework implemented classifies each transmission as either *tactical* or *administrative* traffic. Tactical traffic prioritizes low latency 5G communication while administrative traffic prefers the reliability of SATCOM.

To evaluate resilience, the simulation introduces fault conditions including periodic outages of 5G and SATCOM links. These outages force the SDN controller to reroute traffic with the next best option, enabling the analysis of network

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adaptability under degraded conditions. Performance of the network is evaluated using *delivery success rate*, *latency*, and *overall reliability*.

By simulating these interactions, the project aims to answer a core question: *How effectively can SDN routing strategies between 5G and SATCOM improve communication resilience and reliability within a CSG under degraded network conditions?*

To address this question, we build on existing SDN maritime architectures that propose multi controller and "ship-as-switch" designs [1], [3], [9], and extend them into a fleet-level MATLAB model [11], [12]. The key contributions of this work are:

- A multi-band, multilink communication model for a CSG that integrates communication across the 5G and SATCOM spectrum.
- An SDN based control framework that prioritizes traffic based on mission criticality and the available paths to the intended destination.
- A quantitative performance evaluation using packet level reliability and latency metrics.
- Comprehensive performance evaluation across a diverse set of contested operational and environmental scenarios–including degraded SATCOM and varying Line-Of-Sight (LOS) conditions, providing quantitative metrics on the reliability and resilience achieved by various link selection strategies.

A. List of Acronyms

TABLE I
NOMENCLATURE AND ACRONYMS

Acronym	Definition
AES	Advanced Encryption Standard
C2	Command and Control
CSG	Carrier Strike Group
CVN	Aircraft Carrier (Nuclear-powered)
GCM	Galois/Counter Mode
gNodeB	Next Generation NodeB (5G Base Station)
HF	High Frequency
LOS	Line-Of-Sight
MATLAB	Matrix Laboratory
mmWave	Millimeter Wave
SATCOM	Satellite Communications
SDN	Software Defined Networking
SDR	Software Defined Radio
SINR	Signal-to-Interference-plus-Noise Ratio
UE	User Equipment
UHF	Ultra High Frequency
VHF	Very High Frequency

The remainder of this paper is organized as follows. Section II reviews relevant literature on SDN, followed by their integration in maritime and tactical networks in Section III. Section IV describes the simulation architecture, experimental design, and performance results, followed by conclusions and directions for future research in Section V.

II. RELATED WORK

A. Foundations of Software Defined Networking

Software Defined Networking (SDN) represents a major shift from traditional network architectures by decoupling the control and data planes [1]. The separation allows network administrators to centrally manage network behavior through programmable controllers rather than manually configuring each device [2]. SDN provides three key advantages that are directly relevant to naval operations: centralized control, network wide visibility, and dynamic adaptability [3]. These features enable rapid reconfiguration of communication flows, bandwidth allocation, and quality of service management. For complex distributed systems such as Carrier Strike Groups, SDN's logically centralized yet physically distributed control structure improves coordination and reduces latency by optimizing routing decisions across multiple platforms and communication links [4].

B. Hybrid 5G and SATCOM Link Architectures

Modern maritime networks are moving away from fixed-waveform legacy radios toward hybrid architectures that leverage the high throughput of 5G (millimeter wave and Sub-6GHz) and the global reach of Geostationary SATCOM [1], [7]. Unlike previous generations of maritime communication, these hybrid systems use distance-dependent spectral efficiency to select between *High*, *Mid*, and *Low* 5G bands based on the physical topology of the fleet [10]. In this model, 5G serves as the primary low-latency conduit for tactical data, while SATCOM provides a resilient, high-altitude backhaul for administrative traffic or long-range failover [7]. The ability to switch between these bands based on distance and signal interference is fundamental to maintaining a connected fleet in diverse operational theaters.

C. Priority-Based Routing and Autonomous Failover

The integration of SDN logic into hybrid networks allows for Priority-Based Routing, where the network automatically classifies traffic based on mission-criticality [8], [9]. Literature suggests that tactical data requires strict latency bounds often met only by 5G, whereas administrative data can tolerate the propagation delays inherent in SATCOM [11], [9].

The main challenge in these environments is maintaining connectivity during intermittent link outages. By implementing Independent Periodic Fault models, which simulate 5G fading or SATCOM blockage, the SDN controller can execute autonomous failovers [7], [12]. This ensures that "Tactical" packets are rerouted to SATCOM as a secondary path, maintaining reliability even when the primary high-speed link is compromised [9].

D. Simulation Frameworks for Resilient Networks

Simulation provides a testbed for evaluating SDN policies before deployment in high-stakes naval environments [10]. Discrete-event models, particularly those implemented in MATLAB, allow for the precise calculation of geographic

constraints using the Haversine formula to model real-world signal reach [10], [12].

Recent research emphasizes the importance of modeling both propagation delay and transmission time to accurately capture the performance of SATCOM links ($\approx$ 0.23 per hop) versus 5G links (¡100ms) [1], [5]. These frameworks allow researchers to observe how centralized decision-making improves network resilience under randomized loss scenarios and variable bandwidth allocations [11].

E. Security and Cryptographic SDN Logic

While SDN centralizes control, it also creates a focal point for security logic. Modern SDN implementations incorporate Adaptive Encryption, where the controller applies cryptographic protocols based on the security posture of the chosen link [13]. For instance, when the SDN controller forces tactical traffic onto an unencrypted SATCOM relay due to 5G unavailability, it can dynamically trigger AES-256-GCM encryption to ensure data integrity and confidentiality [8], [9]. This approach is critical in contested maritime environments where communications are subject to both kinetic and electronic interference [1].

III. METHODOLOGY

This section details the simulation of an SDN-controlled hybrid communication network for a Carrier Strike Group (CSG) using MATLAB. The framework transitions away from legacy radio abstractions to focus on a high fidelity interplay between multi-band 5G and Geostationary SATCOM under a centralized software-defined controller [1], [2]. The methodology emphasizes a programmable network state, allowing for reproducible analysis of latency and reliability in dynamic maritime environments [7], [10].

A. Simulation Environment and Network Modeling

The simulation utilizes a discrete-event approach to model eight primary nodes: five guided-missile ships, an aircraft carrier (CVN), a shore-based command facility, and a geostationary satellite.

Spatial Geometry: Node positions are defined via geographic coordinates, with relative distances calculated using the Haversine formula to account for Earth's curvature [10]. This is essential for determining the physical eligibility of range-limited 5G bands.

$$\alpha = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right) \tag{1}$$

$$d = 2R \cdot \text{atan2}(\sqrt{\alpha}, \sqrt{1-\alpha}) \tag{2}$$

SDN-Enabled Nodes: The carrier is modeled as the primary 5G gNodeB and the logical host for the SDN Controller [1], [4]. Ships are treated as User Equipment (UE) that dynamically request bandwidth based on mission requirements and their proximity to the carrier [7], [12].

B. Hybrid Link and Channel Modeling

The network employs a tiered link architecture that shifts traffic between high capacity 5G and high coverage SATCOM based on real-time link health and geographic constraints.

1. Multi-Band 5G Model

The 5G component is abstracted into three operational bands determined by the distance d between communicating nodes:

- High-band (mmWave): $d \leq 1.6$ km providing 1 Gbps throughput.
- Mid-band (Sub-6GHz): $1.6 < d \leq 20$ km providing 100 Mbps throughput.
- Low-band (Sub-1GHz): $20 < d \leq 55$ km providing 10 Mbps throughput.

2. SATCOM Model

SATCOM is utilized as a globally available but high-latency relay for administrative traffic or tactical failover [1]. The total latency τ is calculated as the sum of propagation time, transmission time, and retransmission penalties [10]:

$$\tau = \frac{n \cdot 2h}{c} + \frac{L}{R} + \text{Retries} \cdot \text{Penalty} \tag{3}$$

Where h is the satellite altitude (35,000 km), L is the payload size, and R is the variable data rate (2, 4, or 8 Mbps).

To account for dynamic channel conditions, the simulation introduces a stochastic interference variable, r. This variable represents the aggregate impact of environmental factors such as atmospheric attenuation or signal fading [10]. The SDN controller compares r against pre-defined reliability thresholds (e.g., 0.35 for High-band 5G) to determine if a link meets the Signal-to-Interference-plus-Noise Ratio (SINR) requirements for tactical data transmission [7], [9].

C. SDN Control Logic and Fault Resilience

The simulation implements a centralized control plane co-located with the carrier, which maintains a global view of the network's state [2], [3]. The controller executes a per-packet routing policy:

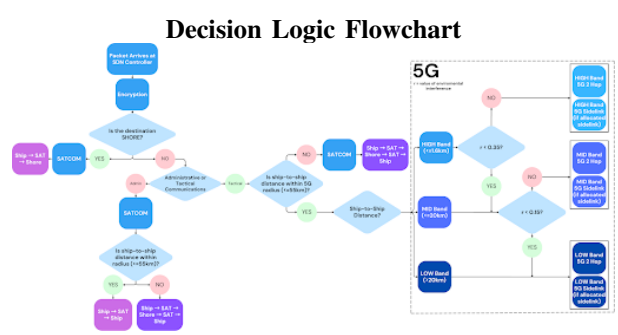

Fig. 2. SDN Controller Decision Logic for Priority-Based Path Selection and 5G Band Allocation

The branching logic for multi-band selection and encryption triggers is visualized in Figure 2, illustrating the automated failover from range-limited 5G to high-latency SATCOM [1], [7].

Priority-Based Routing: Packets are tagged as Tactical or Administrative. Tactical packets are prioritized for low-latency 5G links. If forced onto SATCOM due to range or link failure, the controller applies AES-256-GCM encryption to ensure data integrity [13].

Independent Periodic Faults: To simulate real-world interference or hardware failures, the network injects independent faults. SATCOM links experience outages every 10th packet, while 5G links experience outages every 8th packet.

Failover Mechanism: Upon detecting a link fault or out-of-range condition, the SDN controller automatically recalculates the flow path to the secondary link [7], [9]. A complete packet loss is only recorded if both primary and secondary paths are simultaneously unavailable.

D. Data Collection and Metrics

Quantitative outputs are captured for 100 discrete communication events to evaluate the following [2], [10]:

- **End-to-End Latency:** The temporal cost of routing, transmission, and retries.
- **Reliability:** The percentage of successful deliveries under periodic fault conditions.
- **Path Selection Efficiency:** The ratio of 5G vs. SATCOM utilization based on the CSG's spatial distribution.

IV. RESULTS

Performance was evaluated across 100 simulated events under various distances and link failures.

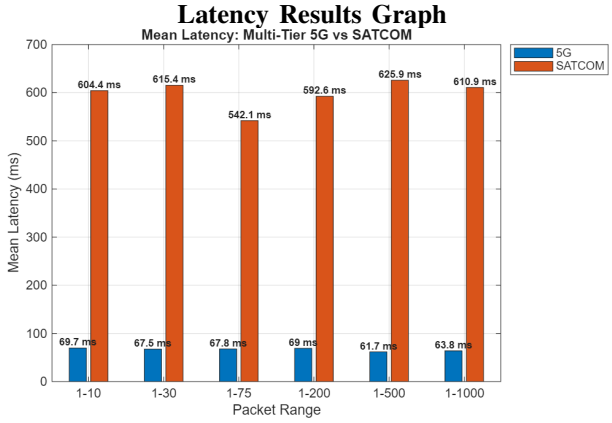

Fig. 3. Mean Latency for 5G and SATCOM

The key results are as follows:

- 5G Latency: ∼61-86 ms
- SATCOM Latency: ∼483-622 ms
- Performance Gap: SATCOM is ∼6-10x slower than 5G

5G consistently maintains low latency due to high throughput and short range communication. Band selection based on distance allows efficient resource utilization across different ranges. SATCOM exhibits higher latency due to long propagation distances and retransmission effects through a satellite and a shore node. Latency increases further in multi hop paths.

TABLE II
PACKET RESULT SAMPLES

Event ID	Path	Link	Security	Latency
PKT 040	Ship5-CVN	Down	N/A	FAILED
PKT 041	CVN-Shore	SATCOM	AES-256	0.733 s
PKT 042	Shore-Ship3	SATCOM	AES-256	2.483 s
PKT 043	CVN-Ship4	5G (Low)	Built-In	0.100 s

The SDN controller successfully reroutes traffic during link failures. Packet loss only occurs when both links are unavailable. Despite these periodic failures, the network maintains high reliability across the 100 events, which demonstrates the effectiveness of a SDN based framework in degraded environments. This framework provides an operational advantage by dynamically selecting the most optimal communication path under changing conditions.

V. CONCLUSION

This project presented a MATLAB based simulation of an SDN enabled framework for a Carrier Strike Group. By integrating both 5G and SATCOM links, results demonstrate that 5G provides lower latency, making it ideal for tactical communications, while SATCOM offers reliable long range connectivity. The SDN controller effectively balances these trade offs through dynamic routing and redundancy. The network maintains high reliability under degrading conditions and periodic link failures. This highlights the importance of a centralized, adaptive network control framework in modern naval operations.

VI. FUTURE WORK

Future improvements from this project include simulating this framework using Simulink to provide detailed SDR waveform modeling. Additionally, scenarios consisting of adversarial jamming and spoofing would add a layer of realism. Lastly, real world data can be implemented to validate the true effects of a SDN based framework in truly operational environments.

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